

Inherited diseases of haemoglobins

Diagnosis and treatment

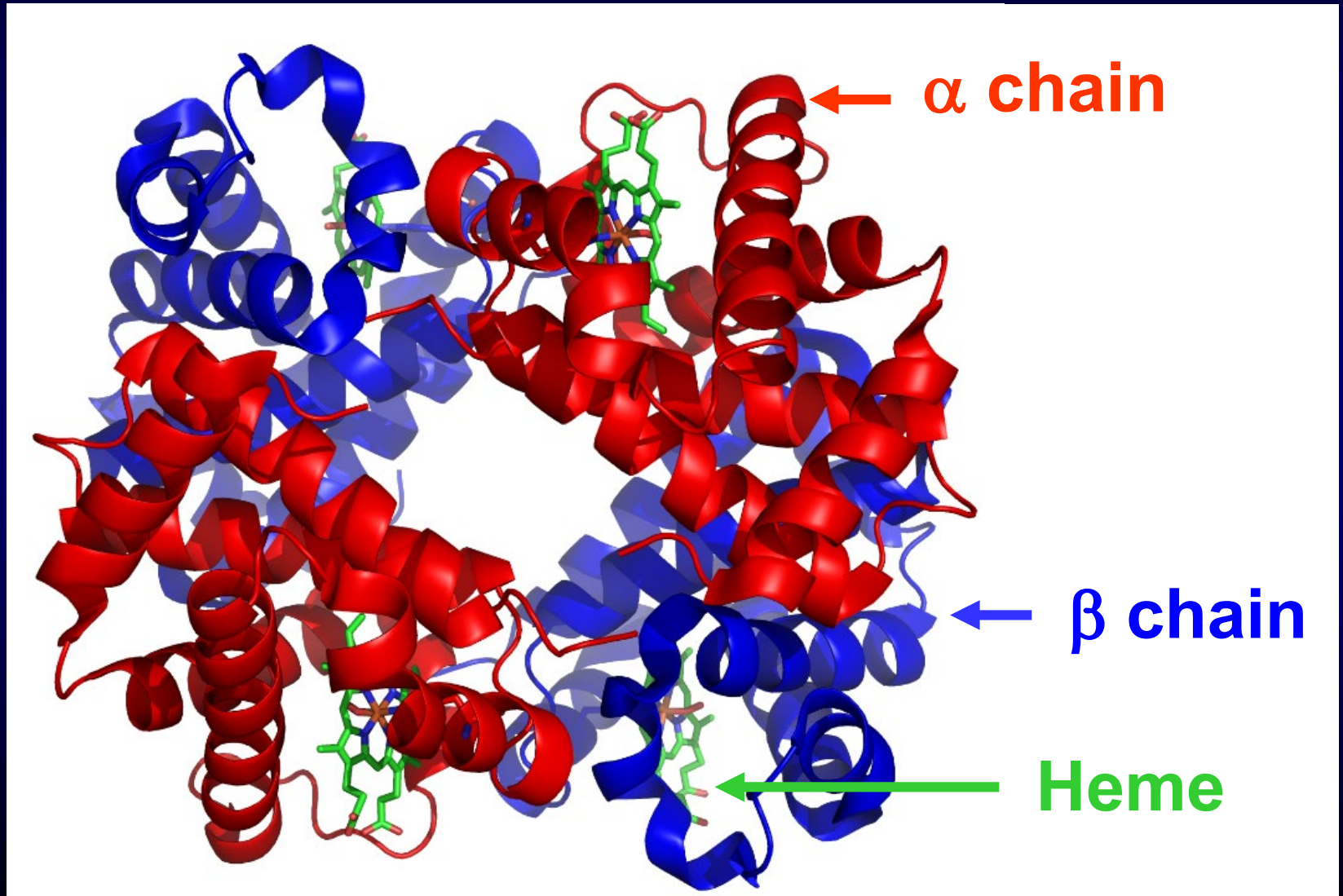
Eric Tse

Department of Medicine

Jason C.C. So

Department of Pathology

Structure of the haemoglobin molecule



Abnormal haemoglobins are caused by mutations of genes encoding the globin chains of the haemoglobin molecule

These gene mutations may result in

- 1. decreased production of globin chains (thalassaemia)**
- 2. globin chains with abnormal function (haemoglobinopathy)**

Hereditary diseases of haemoglobin

1. Thalassaemias

Predominantly a quantitative defect
in the synthesis of globin chains
of the haemoglobin molecule

2. Haemoglobinopathies

Predominantly a qualitative defect
of the production of structurally
abnormal haemoglobin
molecules

1. Decreased production of globin proteins (thalassaemias)

1. Decreased α globin production

α thalassaemia

2. Decreased β globin production

β thalassaemia

2. Abnormal globin structure (haemoglobinopathy) leading to abnormal haemoglobin functions

- 1. Aggregation and reduced solubility**
- 2. Unstable structure**
- 3. Increased oxygen affinity**
- 4. Decreased oxygen affinity**
- 5. Methaemoglobinaemia**

Thalassaemias

α thalassaemia

Due to mutations involving the α globin gene (Chr 16)

There are 4 α globin gene loci

α globin gene is mainly affected by deletions, but mutations may also be found

β thalassaemia

Due to mutations involving the β globin gene (Chr 11)

There are 2 β globin gene loci

β globin gene is mainly affected by mutations, but deletions may also be found

β^0 mutations: no β chain produced

β^+ mutations: β chain production much reduced

Clinical syndromes of thalassaemia

1. Thalassaemia major

α thalassaemia: all 4 α genes deleted

β thalassaemia: both β genes mutated (β^0/β^0 or β^0/β^+)

2. Thalassaemia intermedia

α thalassaemia: 3 α genes deleted

β thalassaemia: both β genes mutated (β^+/ β^+)

3. Thalassaemia trait

α thalassaemia: 1 or 2 α genes deleted

β thalassaemia: 1 β gene mutated (either β^0 or β^+)

β^0 : genetic mutations leading to no production of β globin

β^+ : genetic mutations leading to some production of β globin

Thalassaemia major

α thalassaemia major

hydrops fetalis (Hb Bart's – 4 γ globin)

dies in utero or after birth

may survive if active measures are taken

β thalassaemia major (Cooley's anaemia)

anaemia in first year of life

transfusion dependent for life

Thalassaemia intermedia

α thalassaemia intermedia

(haemoglobin H disease – 4 β globin)

β thalassaemia intermedia

Moderate degree of anaemia

haemoglobin: 6 – 10 g/L

Mild jaundice

Mild to moderate splenomegaly

Thalassaemia trait

α thalassaemia trait

β thalassaemia trait

Mild to no anaemia

Haemoglobin: 10 – 13 g/dL

Mild to no jaundice

Usually no splenomegaly

Thalassaemia intermedia or trait

- 1. Differential diagnosis of hypochromic microcytic anaemia**
- 2. Differential diagnosis of mild splenomegaly**
- 3. Increased ferritin in some cases**
- 4. Family screening is needed when one member is diagnosed to have thalassaemia**
- 5. For at-risk couples, pre-natal diagnosis is needed when pregnancy occurs**
- 6. When the anaemia is more severe than what one expects from the thalassaemia, investigations for other causes of anaemia are necessary**

Differential diagnosis between thalassaemia trait and iron deficiency anaemia

	thalassaemia trait	iron deficiency
Haemoglobin	10 – 13 g/dL	any level
RBC count	normal or ↑	decreased
MCV	↓ but usually not < 65 fl	any level
Serum iron	normal	decreased
TIBC	normal	increased
% iron saturation	normal	decreased

Haemoglobinopathies

1. Aggregation and reduced solubility

Haemoglobin S

Sickle cell

Occurs in people with black ancestry

Not found in the Chinese

Sickling of red blood cells when oxygen
is decreased

Haemoglobinopathies

2. Unstable

Haemoglobin Köln

Unstable haemoglobin chains that unfold and precipitate

Precipitated haemoglobin: Heinz body

Haemolytic anaemia, which may be triggered and exacerbated by infective episodes

Haemoglobinopathies

3. Increased oxygen affinity

Less likely to release the bound oxygen

Causes erythrocytosis

4. Decreased oxygen affinity

P_{50} will be decreased, with patient relatively asymptomatic

5. Methaemoglobinaemia

Mutations lead to the iron in the haeme ring being maintained at a ferric state

Methaemoglobin is blue in colour

Methaemoglobin is increased in the presence of oxidizing agents

TELL US WHAT

SC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)